

Recursive Factorization Systems and Pratt Forests

A unified theory for natural numbers, Pratt polynomials, general polynomial factorization, and hierarchical dependency systems

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with mathematical assistance from ChatGPT 5.6

Status. This is a synthesis and generalization of the uploaded research notes on numerical and polynomial Pratt coordinates, complete fronts, Hadamard–Pratt arithmetic, and root-label fields. The abstract theorems are proved here from the stated axioms. The software interpretation is a mathematical modeling proposal; no empirical performance claim is made.

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Abstract

The common mechanism behind Pratt trees of integers, the recursively defined prime-index polynomials $f_p(x)$, and the polynomial predecessor $D\varphi = \varphi - \varphi(0)$ is not primality itself. It is a ranked factorization rule on atomic objects. Each atom has a predecessor that factors into lower-rank atoms; this rule generates a finite rooted tree. A general object factors into atoms and therefore generates a forest. The total number of occurrences of each atom gives recursive coordinates.

We first develop the polynomial prototype in detail. We then define a *ranked recursive factorization system*: a reduced factorial commutative monoid together with a rank-decreasing predecessor map on its atoms. We prove finite recursion, incidence and path formulas, boundary inversion, intrinsic characterization of the natural coordinate cone, telescoping reconstruction in the quotient group, Hilbert-space embeddings, transport of linear functionals, Hadamard closure, and an induced semiring. We add a second, additive layer: recursive realizations in commutative rings or semirings. This layer yields complete-front expansions, multivariate master polynomials, probability distributions, cumulants, conservation laws, and general tree inequalities.

Natural numbers, general monic polynomials, and the prime-index Pratt polynomials are recovered as specializations. Finally, package-version dependency systems are modeled as a free factorial monoid whose predecessor of a package is its direct dependency multiset. This gives exact formulas for unfolded dependency counts, root reconstruction, consistency tests, similarity kernels, overlap products, and checkpoint-front polynomials, while clearly separating a dependency DAG from its occurrence-unfolded forest.

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1 Introduction: atoms, roots, and predecessors

The word *root* has two different meanings that must not be conflated.

- An *atom type* is an irreducible building block: a prime number, a monic irreducible polynomial, or a package-version identifier.
- A *root occurrence* is an occurrence of an atom at the top level of the factorization of a particular object.

The same atom type can occur as a root in one forest and as an internal node or leaf in another. For example, in the polynomial forest of

$$f = (x + 1)(x^2 + 1),$$

the atoms $x + 1$ and $x^2 + 1$ label roots, while x labels descendants. All three are irreducible.

The central principle is the following.

Recursive factorization principle. An atomic object α is equipped with a predecessor $D\alpha$. The factorization of $D\alpha$ determines the children of an α -node. A rank function guarantees that every child has smaller rank, so the recursion terminates. A general object is a product of atoms and therefore gives a disjoint union of atomic trees.

There are two logically distinct layers.

Layer 1. Multiplicative-combinatorial layer. Factorization, trees, node-count coordinates, the transition matrix, boundary inversion, and telescoping reconstruction.

Layer 2. Additive realization layer. Identities of the form

$$g_\alpha = c_\alpha + \prod_{\beta} g_\beta^{B_{\beta,\alpha}},$$

which expand as weighted sums over complete fronts.

Natural numbers and polynomials possess both layers. Software dependencies possess the first layer canonically; the second can be supplied by choosing counting, cost, risk, or reliability semirings.

2 The polynomial prototype

2.1 The factorial monoid of monic polynomials

Let

$$\mathcal{I} = \{\varphi \in \mathbb{Q}[x] : \varphi \text{ is nonconstant, monic, and irreducible}\}.$$

Let $\mathcal{M}_{\text{poly}}$ be the multiplicative monoid of monic polynomials in $\mathbb{Q}[x]$, including 1. Unique factorization gives

$$f = \prod_{\varphi \in \mathcal{I}} \varphi^{v_\varphi(f)}, \quad v_\varphi(f) \in \mathbb{N}_0,$$

with finite support.

2.2 The predecessor operator

Definition 2.1 (Polynomial predecessor and local ratio). For $\varphi \in \mathcal{I}$, define

$$D\varphi = \begin{cases} 1, & \varphi = x, \\ \varphi(x) - \varphi(0), & \varphi \neq x, \end{cases} \quad \Delta(\varphi) = \frac{\varphi}{D\varphi} \in \mathbb{Q}(x)^\times.$$

The convention $Dx = 1$ is essential: the naive rule $Dx = x - x(0) = x$ would create an infinite loop.

Lemma 2.2 (Degree descent). *If $\varphi \neq x$, then $D\varphi$ is monic and divisible by x . If $\deg \varphi = 1$, then $D\varphi = x$. If $\deg \varphi \geq 2$, every irreducible factor of $D\varphi$ has degree strictly smaller than $\deg \varphi$.*

Proof. An irreducible $\varphi \neq x$ has $\varphi(0) \neq 0$. Subtracting the constant term preserves the leading coefficient and makes the constant term zero, hence $x \mid D\varphi$. If $\deg \varphi = 1$, monicity forces $D\varphi = x$. If $\deg \varphi = d \geq 2$, the factor x consumes positive degree, so no irreducible factor of $D\varphi$ can have degree d . $\square$

Factor

$$D\varphi = \prod_{\psi \in \mathcal{I}} \psi^{B_{\psi,\varphi}}, \quad B_{\psi,\varphi} := v_\psi(D\varphi).$$

The matrix B is strictly rank-decreasing with respect to polynomial degree.

2.3 Polynomial Pratt trees and forests

Definition 2.3 (Polynomial recursive tree). The tree T_φ has a root labelled φ . For each $\psi \in \mathcal{I}$, the root has $B_{\psi,\varphi}$ children labelled ψ , and below each such child lies a copy of T_ψ . The terminal tree T_x is a single vertex.

Definition 2.4 (Polynomial forest). For

$$f = \prod_{\varphi} \varphi^{v_\varphi(f)},$$

define

$$F_f := \bigsqcup_{\varphi \in \mathcal{I}} v_\varphi(f) T_\varphi.$$

For $\psi \in \mathcal{I}$, let $M_\psi(f)$ be the total number of ψ -labelled vertices in F_f .

Equivalently,

$$M_\psi(\varphi) = \mathbf{1}_{\psi=\varphi} + \sum_{\eta \in \mathcal{I}} v_\eta(D\varphi) M_\psi(\eta),$$

and

$$M_\psi(f) = \sum_{\varphi \in \mathcal{I}} v_\varphi(f) M_\psi(\varphi).$$

Degree descent proves well-definedness and finite support.

Proposition 2.5 (Complete additivity). *For monic $f, g \in \mathbb{Q}[x]$,*

$$M_\psi(fg) = M_\psi(f) + M_\psi(g) \quad (\psi \in \mathcal{I}).$$

Proof. The forest F_{fg} is the disjoint union $F_f \sqcup F_g$, with root multiplicities added by unique factorization. $\square$

2.4 Boundary inversion

Write

$$M(f) = (M_\psi(f))_{\psi \in \mathcal{I}}, \quad v(f) = (v_\psi(f))_{\psi \in \mathcal{I}}.$$

Every ψ -node is either a root or a child of a higher-rank node. Hence

$$M(f) = v(f) + BM(f).$$

Theorem 2.6 (Polynomial boundary inversion). *Define*

$$\beta := I - B.$$

Then

$$\boxed{\beta M(f) = v(f).}$$

Componentwise,

$$\boxed{v_\psi(f) = M_\psi(f) - \sum_{\varphi \in \mathcal{I}} v_\psi(D\varphi) M_\varphi(f).}$$

Proof. The sum counts exactly the non-root ψ -nodes: every φ -node contributes $v_\psi(D\varphi)$ immediate ψ -children. Subtracting them from all ψ -nodes leaves the root multiplicity. $\square$

2.5 Telescoping reconstruction

Theorem 2.7 (Polynomial Pratt reconstruction). *Every monic $f \in \mathbb{Q}[x]$ satisfies*

$$\boxed{f(x) = \prod_{\varphi \in \mathcal{I}} \left(\frac{\varphi(x)}{D\varphi(x)} \right)^{M_\varphi(f)}.$$

Only finitely many factors differ from 1.

Proof. The exponent of a fixed irreducible ψ on the right is

$$M_\psi(f) - \sum_{\varphi} v_\psi(D\varphi) M_\varphi(f) = v_\psi(f)$$

by boundary inversion. Thus the product has exactly the irreducible factorization of f . $\square$

For a nonzero, not necessarily monic polynomial $f = \lambda f_0$, with $\lambda = \text{lc}(f)$ and f_0 monic,

$$f = \lambda \prod_{\varphi \in \mathcal{I}} \left(\frac{\varphi}{D\varphi} \right)^{M_\varphi(f_0)}.$$

2.6 A new complete-front expansion for general polynomials

The predecessor identity is additive as well as multiplicative:

$$\varphi = \varphi(0) + D\varphi.$$

This gives a complete-front expansion not only for the special polynomials f_n , but for every monic rational polynomial.

For a nonterminal irreducible $\varphi \neq x$, assign the stop weight

$$w(\varphi) = \varphi(0),$$

and assign the terminal weight $w(x) = x$. For a complete front A of T_φ , put

$$w(A) = \prod_{v \in A} w(\ell(v)).$$

Theorem 2.8 (General polynomial front expansion). *For every $\varphi \in \mathcal{I}$,*

$$\varphi(x) = \sum_{A \in \text{CF}(T_\varphi)} \prod_{v \in A} w(\ell(v)).$$

More generally, for every monic f ,

$$f(x) = \sum_{A \in \text{CF}(F_f)} \prod_{v \in A} w(\ell(v)).$$

Proof. For $\varphi = x$, the unique front is the terminal vertex and the formula gives x . Let $\varphi \neq x$. A complete front either stops at the root, contributing $\varphi(0)$, or avoids the root and independently chooses a complete front in every child subtree. By induction on degree, the expanded contribution is

$$\prod_{\psi \in \mathcal{I}} \psi(x)^{v_\psi(D\varphi)} = D\varphi(x).$$

Thus the total is $\varphi(0) + D\varphi = \varphi$. Forest components multiply, and expanding the product is equivalent to choosing one complete front in every component. $\square$

Example 2.9. For $f = (x + 1)(x^2 + 1)$, the two root trees have front sums

$$x + 1, \quad x^2 + 1.$$

The forest fronts therefore give

$$f = 1 + x + x^2 + x^3.$$

The four terms correspond to choosing the root or descending to the terminal x -nodes independently in the two components.

2.7 Differential consequences in the polynomial realization

Define the boundary radical

$$\text{rad}_D(f) := \prod_{\theta \in \mathcal{I}: \beta_\theta(M(f)) > 0} \theta.$$

Boundary inversion gives $\text{rad}_D(f) = \text{rad}(f)$. Taking logarithmic derivatives in the reconstruction formula yields

$$\frac{f'}{f} = \sum_{\varphi \in \mathcal{I}} M_\varphi(f) \left(\frac{\varphi'}{\varphi} - \frac{(D\varphi)'}{D\varphi} \right) = \sum_{\theta \in \mathcal{I}} v_\theta(f) \frac{\theta'}{\theta}.$$

Consequently,

$$\frac{f}{\text{rad}(f)} \mid f'.$$

If $a + b = c$ are pairwise coprime, nonzero rational polynomials and are not all constant, the Wronskian argument gives Mason–Stothers:

$$\max\{\deg a, \deg b, \deg c\} \leq \deg \text{rad}(abc) - 1.$$

The terminal coordinate is exactly degree:

$$M_x(f) = \deg f.$$

For every irreducible ψ one has the subtree-capacity bound

$$(\deg \psi) M_\psi(f) \leq \deg f.$$

Indeed, every occurrence of ψ roots a disjoint descendant subtree containing exactly $\deg \psi$ terminal x -nodes.

3 Abstract ranked recursive factorization systems

3.1 The multiplicative axioms

Definition 3.1 (Reduced factorial monoid). A commutative cancellative monoid H is *reduced factorial* if its only unit is 1 and every $h \in H$ has a unique finite factorization

$$h = \prod_{\alpha \in \mathcal{A}} \alpha^{v_\alpha(h)}$$

into a fixed set $\mathcal{A}$ of atom representatives.

Thus $H \cong \mathbb{N}_0^{(\mathcal{A})}$ as a commutative monoid, although we retain multiplicative notation.

Definition 3.2 (Ranked recursive factorization system). A ranked recursive factorization system is a quadruple

$$\mathfrak{R} = (H, \mathcal{A}, D, \rho)$$

where:

- (i) H is a reduced factorial commutative monoid with atoms $\mathcal{A}$;
- (ii) $D : \mathcal{A} \rightarrow H$ is a predecessor map;
- (iii) $\rho : \mathcal{A} \rightarrow \mathbb{N}_0$ is a rank function;
- (iv) if $\beta \mid D\alpha$, then $\rho(\beta) < \rho(\alpha)$.

An atom τ is terminal if $D\tau = 1$. The terminal set is denoted $\mathcal{T}$.

Write

$$D\alpha = \prod_{\beta \in \mathcal{A}} \beta^{B_{\beta,\alpha}}, \quad B_{\beta,\alpha} = v_\beta(D\alpha) \in \mathbb{N}_0.$$

Every column of B has finite support. Rank descent implies local nilpotence: for every finitely supported vector x , $B^k x = 0$ for all sufficiently large k .

3.2 Atomic trees and object forests

Definition 3.3. The tree T_α has root label α and $B_{\beta,\alpha}$ children of type β , each carrying a copy of T_β . For

$$h = \prod_{\alpha} \alpha^{v_\alpha(h)},$$

define

$$F_h = \bigsqcup_{\alpha} v_\alpha(h) T_\alpha.$$

The recursive coordinate $M_\beta(h)$ is the total number of β -labelled vertices in F_h .

Theorem 3.4 (Finiteness). *Every T_α and every F_h is finite, and $M(h)$ has finite support.*

Proof. Ranks strictly decrease along every edge. Since ranks are nonnegative integers, every path has length at most $\rho(\alpha)$, and finite branching then gives a finite tree. $\square$

3.3 Incidence matrix, resolvent, and path formula

Define

$$C_{\beta, \alpha} := M_{\beta}(\alpha).$$

The α -column of C is the full population vector of T_{α} .

Theorem 3.5 (Resolvent identity). *On every finite rank-closed truncation,*

$$C = I + CB = (I - B)^{-1} = I + B + B^2 + \dots.$$

Consequently,

$$M(h) = Cv(h).$$

Proof. The root of T_{α} contributes e_{α} . Every immediate γ -child roots a copy of T_{γ} , and there are $B_{\gamma, \alpha}$ such children. Hence

$$C_{\beta, \alpha} = \delta_{\beta, \alpha} + \sum_{\gamma} C_{\beta, \gamma} B_{\gamma, \alpha},$$

which is $C = I + CB$. Local nilpotence gives the finite geometric inverse. Summing the columns with root multiplicities $v_{\alpha}(h)$ gives $M(h) = Cv(h)$. $\square$

Corollary 3.6 (Weighted path expansion).

$$C_{\beta, \alpha} = \sum_{k \geq 0} \sum_{\alpha = \alpha_0 \rightarrow \alpha_1 \rightarrow \dots \rightarrow \alpha_k = \beta} \prod_{j=1}^k B_{\alpha_j, \alpha_{j-1}}.$$

The $k = 0$ term is 1 when $\alpha = \beta$ and 0 otherwise.

Thus $C_{\beta, \alpha}$ counts descending dependency paths with multiplicity.

3.4 Boundary inversion and the natural cone

Theorem 3.7 (Abstract boundary theorem). *For every $h \in H$,*

$$v(h) = (I - B)M(h).$$

Proof. This is the inverse relation to $M(h) = Cv(h)$, since $C^{-1} = I - B$. Combinatorially it subtracts all forced child occurrences from the total population, leaving root occurrences. $\square$

Define the natural recursive cone

$$\mathcal{C}_D := M(H) \subseteq \mathbb{N}_0^{(\mathcal{A})}.$$

Theorem 3.8 (Intrinsic cone characterization). *For a finitely supported integer vector x , the following are equivalent:*

- (i) $x = M(h)$ for some $h \in H$;
- (ii) $x \geq 0$ and $(I - B)x \geq 0$ coordinatewise;
- (iii) $x = (I - B)^{-1}u$ for some $u \in \mathbb{N}_0^{(\mathcal{A})}$.

In that case $u = (I - B)x = v(h)$ and

$$h = \prod_{\alpha} \alpha^{u_{\alpha}}.$$

Proof. Boundary inversion gives (i) $\Rightarrow$ (ii). Local nilpotence and nonnegativity of B give (ii) $\Leftrightarrow$ (iii). Given (iii), define $h = \prod \alpha^{\mu_\alpha}$. Then

$$(I - B)M(h) = v(h) = u = (I - B)x,$$

and injectivity of $I - B$ on finite vectors gives $M(h) = x$. $\square$

Equivalently,

$$\mathcal{C}_D = \{x \in \mathbb{N}_0^{(\mathcal{A})} : x_\beta \geq \sum_{\alpha} B_{\beta, \alpha} x_\alpha \text{ for every } \beta\}.$$

3.5 Telescoping reconstruction in the quotient group

Let $q(H)$ be the quotient group of the cancellative monoid H . Define

$$\Delta(\alpha) := \alpha(D\alpha)^{-1} \in q(H).$$

Theorem 3.9 (Abstract multiplicative reconstruction). *For every $h \in H$,*

$$h = \prod_{\alpha \in \mathcal{A}} \Delta(\alpha)^{M_\alpha(h)} = \prod_{\alpha \in \mathcal{A}} \left(\frac{\alpha}{D\alpha} \right)^{M_\alpha(h)} \quad \text{in } q(H).$$

Proof. The exponent of β on the right is

$$M_\beta(h) - \sum_{\alpha} B_{\beta, \alpha} M_\alpha(h) = ((I - B)M(h))_\beta = v_\beta(h).$$

$\square$

3.6 Hilbert-space coordinates and kernels

Define

$$\Phi_D : H \longrightarrow \ell^2(\mathcal{A}), \quad \Phi_D(h) = M(h).$$

Every image vector has finite support.

Theorem 3.10 (Sparse Hilbert embedding).

$$\Phi_D(hk) = \Phi_D(h) + \Phi_D(k),$$

and Φ_D is injective.

Proof. Forests concatenate under multiplication, giving additivity. Boundary inversion reconstructs $v(h)$ from $M(h)$ and hence reconstructs h . $\square$

For weights $w_\alpha > 0$, define

$$K_w(h, k) = \sum_{\alpha} w_\alpha M_\alpha(h) M_\alpha(k),$$

$$\|h\|_{D, w}^2 = K_w(h, h), \quad d_{D, w}(h, k)^2 = \sum_{\alpha} w_\alpha (M_\alpha(h) - M_\alpha(k))^2.$$

Then K_w is positive semidefinite and $d_{D, w}$ is a metric. Moreover,

$$d_{D, w}(hg, kg) = d_{D, w}(h, k),$$

so common multiplication acts by translation in feature space.

3.7 Transport of root-level linear functionals

Let $a = (a_\alpha)$ be finitely supported and define a root-level score

$$E_{\text{root}}(h) = a^T v(h).$$

Since $v = (I - B)M$,

$$E_{\text{root}}(h) = \eta^T M(h), \quad \eta = (I - B)^T a.$$

Componentwise,

$$\eta_\beta = a_\beta - \sum_\gamma B_{\gamma, \beta} a_\gamma.$$

Thus a score assigned to roots can be transported to a signed score on all recursive occurrences.

4 Recursive realizations and complete fronts

4.1 Additive realizations in a semiring or ring

The multiplicative theory does not require addition. Complete-front expansions do.

Definition 4.1 (Recursive realization). Let R be a commutative semiring or ring. Choose terminal values $t_\tau \in R$ for $\tau \in \mathcal{T}$ and stop weights $c_\alpha \in R$ for nonterminal α . Define recursively

$$g_\tau = t_\tau, \quad g_\alpha = c_\alpha + \prod_{\beta \in \mathcal{A}} g_\beta^{B_{\beta, \alpha}} \quad (\alpha \notin \mathcal{T}).$$

For $h \in H$, define

$$g_h := \prod_\alpha g_\alpha^{v_\alpha(h)}.$$

Rank descent makes the definition finite.

4.2 Complete fronts

A node front of a finite rooted forest is an antichain in the ancestor order. It is complete if it meets every root-to-leaf path. Equivalently, it meets every such path exactly once, or it is maximal among antichains.

Assign to a vertex v the weight

$$w(v) = \begin{cases} t_{\ell(v)}, & v \text{ is a leaf,} \\ c_{\ell(v)}, & v \text{ is nonterminal.} \end{cases}$$

Theorem 4.2 (Universal complete-front expansion). *For every atom α and every $h \in H$,*

$$g_\alpha = \sum_{A \in \text{CF}(T_\alpha)} \prod_{v \in A} w(v),$$

$$g_h = \sum_{A \in \text{CF}(F_h)} \prod_{v \in A} w(v).$$

Proof. A complete front of a nontrivial tree either stops at the root or independently chooses a complete front in every child subtree. This gives precisely the recursion for g_α . Disjoint forest components multiply. $\square$

This theorem simultaneously contains:

- $p = 1 + (p - 1)$ for natural primes, with terminal value 2;
- $f_p(x) = 1 + f_{p-1}(x)$ for Pratt polynomials, with terminal value x ;
- $\varphi(x) = \varphi(0) + D\varphi(x)$ for general irreducible polynomials.

4.3 Fully label-marked and root-label polynomials

Introduce variables z_α for atom labels. Define

$$\Phi_\tau(z) = z_\tau, \quad \Phi_\alpha(z) = z_\alpha + \prod_{\beta} \Phi_\beta(z)^{B_{\beta,\alpha}} \quad (\alpha \notin \mathcal{T}),$$

and

$$\Phi_h(z) = \prod_{\alpha} \Phi_\alpha(z)^{v_\alpha(h)}.$$

Then

$$\Phi_h(z) = \sum_{A \in \text{CF}(F_h)} \prod_{v \in A} z_{\ell(v)}.$$

To retain both the component root type and the selected label, introduce $z_{\alpha,\beta}$. In a component rooted at α , a selected β -node receives $z_{\alpha,\beta}$. The master polynomial is

$$\mathfrak{M}_h((z_{\alpha,\beta})) = \prod_{\alpha} \Phi_\alpha((z_{\alpha,\beta})_\beta)^{v_\alpha(h)}.$$

Label-only and root-separated views are obtained by specialization.

4.4 Degree recovery and capacities

Since rank strictly decreases along every edge, two distinct vertices with the same label are incomparable. Therefore all β -vertices form an antichain and can be extended to a complete front.

Theorem 4.3 (Degree recovery).

$$\deg_{z_\beta} \Phi_\alpha = C_{\beta,\alpha}, \quad \deg_{z_\beta} \Phi_h = M_\beta(h),$$

and

$$\deg_{z_{\alpha,\beta}} \mathfrak{M}_h = v_\alpha(h) C_{\beta,\alpha}.$$

The vector $C_{\bullet,\alpha}$ is a coordinatewise capacity vector of the front polynomial. It need not occur as a single monomial, because one front cannot usually select all vertices simultaneously.

4.5 Probability and cumulants

Choose a complete front of F_h uniformly. If $N_\alpha(A)$ counts selected α -vertices, then

$$\frac{\Phi_h(z)}{\Phi_h(\mathbf{1})}$$

is the multivariate probability-generating function. With logarithmic differential operators

$$D_\alpha = z_\alpha \frac{\partial}{\partial z_\alpha},$$

we have

$$\begin{aligned} \mathbb{E}[N_\alpha] &= D_\alpha \log \Phi_h|_{z=\mathbf{1}}, \\ \text{Cov}(N_\alpha, N_\beta) &= D_\alpha D_\beta \log \Phi_h|_{z=\mathbf{1}}. \end{aligned}$$

For any additive statistic X_h whose probability-generating polynomial is multiplicative under disjoint union,

$$X_{hk} \stackrel{d}{=} X_h + X_k$$

with independent summands. Consequently every cumulant is additive:

$$\kappa_r(hk) = \kappa_r(h) + \kappa_r(k) \quad (r \geq 1).$$

5 Hadamard closure and the induced overlap semiring

5.1 A general cone lemma

Lemma 5.1 (Integer Hadamard inequality). *Let $a_j \in \mathbb{N}_0$ have finite support. If*

$$X \geq \sum_j a_j x_j, \quad Y \geq \sum_j a_j y_j$$

for nonnegative x_j, y_j , then

$$XY \geq \sum_j a_j x_j y_j.$$

Proof. Put $U = \sum a_j x_j$ and $V = \sum a_j y_j$. Then $XY \geq UV$, and

$$UV - \sum_j a_j x_j y_j = \sum_{j \neq k} a_j a_k x_j y_k + \sum_j a_j (a_j - 1) x_j y_j \geq 0.$$

The integrality $a_j \in \mathbb{N}_0$ is essential to this short proof. □

5.2 Hadamard closure of every recursive cone

For vectors x, y , write

$$(x \circ y)_\alpha = x_\alpha y_\alpha.$$

Theorem 5.2 (Abstract Hadamard closure). *If $x, y \in \mathcal{C}_D$, then*

$$x \circ y \in \mathcal{C}_D.$$

Proof. Fix β . Since $x, y \in \mathcal{C}_D$,

$$x_\beta \geq \sum_\alpha B_{\beta, \alpha} x_\alpha, \quad y_\beta \geq \sum_\alpha B_{\beta, \alpha} y_\alpha.$$

Apply the lemma with $a_\alpha = B_{\beta, \alpha}$ to obtain

$$x_\beta y_\beta \geq \sum_\alpha B_{\beta, \alpha} x_\alpha y_\alpha.$$

Thus $(I - B)(x \circ y) \geq 0$. □

Definition 5.3 (Overlap product). For $h, k \in H$, define $h \star_D k$ by

$$M(h \star_D k) = M(h) \circ M(k).$$

Existence follows from Hadamard closure and uniqueness from injectivity of M .

The root multiplicities are explicitly

$$v_\beta(h \star_D k) = M_\beta(h) M_\beta(k) - \sum_\alpha B_{\beta, \alpha} M_\alpha(h) M_\alpha(k).$$

They are nonnegative integers.

A positivity certificate is obtained by writing

$$r_\beta = ((I - B)x)_\beta, \quad s_\beta = ((I - B)y)_\beta,$$

$$X_\beta = \sum_\alpha B_{\beta,\alpha} x_\alpha, \quad Y_\beta = \sum_\alpha B_{\beta,\alpha} y_\alpha.$$

Then

$$\begin{aligned} ((I - B)(x \circ y))_\beta &= r_\beta s_\beta + r_\beta Y_\beta + s_\beta X_\beta \\ &\quad + \sum_{\alpha \neq \gamma} B_{\beta,\alpha} B_{\beta,\gamma} x_\alpha y_\gamma + \sum_\alpha B_{\beta,\alpha} (B_{\beta,\alpha} - 1) x_\alpha y_\alpha. \end{aligned}$$

Corollary 5.4 (Support intersection).

$$\text{supp } M(h \star_D k) = \text{supp } M(h) \cap \text{supp } M(k).$$

Theorem 5.5 (Recursive overlap semiring). *Define*

$$h \oplus_D k := hk, \quad h \otimes_D k := h \star_D k.$$

Then H is a commutative semiring without a required multiplicative identity: $\oplus_D$ becomes vector addition and $\otimes_D$ becomes coordinatewise multiplication under M . In particular,

$$h \star_D (kg) = (h \star_D k)(h \star_D g).$$

The monoid unit 1 is the additive zero and is absorbing for $\star_D$. A $\star_D$ -identity exists exactly when the all-ones vector on the atom set is a valid finitely supported recursive coordinate vector. In infinite atom systems this normally fails.

Idempotents are characterized by

$$h \star_D h = h \iff M_\alpha(h) \in \{0, 1\} \text{ for every } \alpha.$$

Such a 0–1 vector must also satisfy the recursive cone inequalities.

6 General tree inequalities and conservation laws

Let F be any finite rooted forest. Write

$$R(F) = \#\{\text{components}\}, \quad L(F) = \#\{\text{leaves}\},$$

$$I(F) = \#\{\text{non-leaf vertices}\}, \quad V(F) = L(F) + I(F),$$

and $A(F) = \#CF(F)$.

Theorem 6.1 (Edge and leaf-excess identities).

$$\#E(F) = V(F) - R(F),$$

$$L(F) = R(F) + \sum_{v \text{ non-leaf}} (d^+(v) - 1).$$

Consequently $L(F) \geq R(F)$; if every non-leaf has at least $d \geq 2$ children, then

$$L(F) \geq R(F) + (d - 1)I(F).$$

Theorem 6.2 (Front size and front count). *For every complete front A ,*

$$R(F) \leq |A| \leq L(F).$$

Moreover,

$$I(F) + 1 \leq A(F) \leq 2^{I(F)}.$$

Proof. A front meets each component, giving the lower size bound. Map each selected vertex to a descendant leaf; incomparable selected vertices have disjoint descendant subtrees, giving the upper size bound. For one tree with child subtrees T_i ,

$$A(T) = 1 + \prod_i A(T_i), \quad I(T) = 1 + \sum_i I(T_i).$$

Induction gives the count bounds, and forest components multiply. $\square$

6.1 Conserved flows across fronts

Definition 6.3. A weight $w : V(F) \rightarrow R$ is front-conserved if every non-leaf vertex satisfies

$$w(v) = \sum_{u \text{ child of } v} w(u).$$

Theorem 6.4 (Cut conservation). *For every complete front A ,*

$$\boxed{\sum_{v \in A} w(v) = \sum_{r \text{ root of } F} w(r).}$$

Proof. Starting from the root front, replace selected non-leaf vertices by all their children. Each refinement preserves total weight, and every complete front is reached by finitely many refinements. $\square$

Let $\lambda_\tau(\alpha) = C_{\tau, \alpha}$ be the number of terminal τ -leaves below an α -node. Then λ_τ is front-conserved, so for every complete front of F_h ,

$$\boxed{\sum_{v \in A} \lambda_\tau(\ell(v)) = M_\tau(h).}$$

Thus the fully label-marked front polynomial is multihomogeneous with respect to terminal-capacity weights.

If transition probabilities $\pi(v, u) > 0$ sum to 1 over the children of each non-leaf and root mass is 1, then the recursively propagated mass satisfies the Kraft-type equality

$$\boxed{\sum_{v \in A} \mu(v) = R(F)}$$

for every complete front.

For a uniformly random complete front with size Y ,

$$\boxed{R(F) \leq \mathbb{E}[Y] \leq L(F),} \quad \boxed{\text{Var}(Y) \leq \frac{(L(F) - R(F))^2}{4}.}$$

7 Three principal specializations

7.1 Natural numbers

Take

$$H = (\mathbb{N}, \cdot), \quad \mathcal{A} = \mathbb{P},$$

with

$$D(2) = 1, \quad D(p) = p - 1 \quad (p > 2),$$

and rank $\rho(p) = p$ or any strictly increasing function of p . Then

$$B_{q,p} = v_q(p - 1).$$

The recursive coordinates are the numerical Pratt exponents $m_q(n)$:

$$m_q(1) = 0, \quad m_q(ab) = m_q(a) + m_q(b),$$

$$m_q(p) = \mathbf{1}_{q=p} + m_q(p - 1).$$

Boundary inversion gives

$$v_q(n) = m_q(n) - \sum_p v_q(p - 1) m_p(n).$$

Reconstruction gives

$$n = \prod_p \left(\frac{p}{p-1} \right)^{m_p(n)} = \prod_p \left(1 - \frac{1}{p} \right)^{-m_p(n)}.$$

The terminal set is $\{2\}$, and $m_2(n)$ is the number of leaves.

The abstract tree identities specialize to

$$m_2(n) = \Omega(n) + \sum_{q \geq 3} m_q(n) (\Omega(q - 1) - 1),$$

$$\Omega(n) \leq |A| \leq m_2(n) \quad (A \in \text{CF}(F_n)),$$

$$1 + \sum_{q \geq 3} m_q(n) \leq \# \text{CF}(F_n) \leq 2^{\sum_{q \geq 3} m_q(n)}.$$

The sharp logarithmic leaf bounds are

$$\log_3 n \leq m_2(n) \leq \log_2 n.$$

The total number of vertices and height satisfy

$$\sum_q m_q(n) \leq n - 1,$$

$$h(F_n) \leq \log_2(n + 1) - 1.$$

The overlap product becomes the Hadamard–Pratt product

$$M(a \star_P b) = M(a) \circ M(b).$$

For example,

$$2 \star_P n = 2^{m_2(n)}.$$

7.2 The prime-index Pratt polynomials

Define

$$f_1(x) = 1, \quad f_2(x) = x,$$

and for odd primes

$$f_p(x) = 1 + \prod_{q|p-1} f_q(x)^{v_q(p-1)}.$$

For

$$n = \prod_p p^{v_p(n)}, \quad f_n(x) = \prod_p f_p(x)^{v_p(n)}.$$

Then

$$f_{mn} = f_m f_n, \quad f_n(2) = n,$$

and every f_p is irreducible over $\mathbb{Q}$. The polynomial predecessor satisfies

$$Df_p = f_p - f_p(0) = f_{p-1}.$$

The polynomial and numerical recursive coordinates agree:

$$M_{f_p}(f_n) = m_p(n).$$

Hence

$$f_n(x) = \prod_p \left(\frac{f_p(x)}{f_{p-1}(x)} \right)^{m_p(n)}.$$

Evaluation at $x = 2$ gives the numerical reconstruction formula.

The complete-front realization uses nonterminal stop weight 1 and terminal value x :

$$f_n(x) = \sum_{A \in \text{CF}(F_n)} x^{N_2(A)}.$$

Therefore

$$f_n(1) = \# \text{CF}(F_n), \quad \deg f_n = m_2(n),$$

and

$$n = \sum_{A \in \text{CF}(F_n)} 2^{N_2(A)}.$$

After normalization,

$$G_n(s) = \frac{f_n(s)}{f_n(1)}$$

is the probability-generating function of the number of selected terminal vertices in a uniformly random complete front. Its mean and variance are

$$\mu(n) = \frac{f'_n(1)}{f_n(1)},$$

$$\sigma^2(n) = \frac{f''_n(1)}{f_n(1)} + \frac{f'_n(1)}{f_n(1)} - \left(\frac{f'_n(1)}{f_n(1)} \right)^2,$$

and all cumulants are completely additive.

For $x \geq 1$,

$$x^{v_2(n)} f_n(1) \leq f_n(x) \leq x^{m_2(n)} f_n(1),$$

with reversed inequalities for $0 < x \leq 1$. Setting $x = 2$ gives

$$\frac{n}{2^{m_2(n)}} \leq f_n(1) \leq \frac{n}{2^{v_2(n)}}.$$

7.3 General monic rational polynomials

This is the prototype of Section 2 viewed as a specialization of the abstract system:

$$H = M_{\text{poly}}, \quad \mathcal{A} = \mathcal{I}, \quad D\varphi = \varphi - \varphi(0), \quad \rho(\varphi) = \deg \varphi,$$

with terminal atom x . The weighted front expansion has stop weights $\varphi(0)$ and terminal value x . The resulting coefficients need not be nonnegative, but the identity remains exact in $\mathbb{Q}[x]$.

8 Software products as recursive factorization objects

8.1 Package-version atoms

Let $\mathcal{A}_{\text{sw}}$ be a set of package-version identifiers, for example

openssl@3.2.1, libcrypto@3.2.1, zlib@1.3.1.

Treat different versions as different atoms. Let

$$H_{\text{sw}} = \mathbb{N}_0^{(\mathcal{A}_{\text{sw}})}$$

in multiplicative notation: a product is a multiset of root packages.

For a package P , let

$$D(P) = \prod_Q Q^{B_{Q,P}}$$

be its direct dependency multiset. Assume initially that the dependency relation is acyclic. A topological rank gives

$$B_{Q,P} > 0 \implies \rho(Q) < \rho(P).$$

Thus $(H_{\text{sw}}, \mathcal{A}_{\text{sw}}, D, \rho)$ is a ranked recursive factorization system.

Interpretation.

- $v_P(S)$: directly declared root multiplicity of package P in product S .
- $M_Q(S)$: total number of occurrences of Q in the fully unfolded dependency forest.
- $B_{Q,P}$: number of direct Q -dependency occurrences required by one P occurrence.
- $C_{Q,P}$: total number of unfolded Q occurrences generated by one root P .

8.2 Dependency explosion and provenance inversion

For a root vector $u = v(S)$,

$$M(S) = Cu = (I - B)^{-1}u.$$

This is the exact dependency-explosion formula. Conversely,

$$u = (I - B)M(S).$$

Thus a total occurrence inventory can be tested for compatibility with the declared dependency rules:

$$m \geq 0, \quad (I - B)m \geq 0, \quad (I - B)m \in \mathbb{Z}^{(\mathcal{A}_{\text{sw}})}.$$

If a coordinate of $(I - B)m$ is negative, the observed inventory is inconsistent with the assumed deterministic dependency model.

The path formula

$$C_{Q,P} = \sum_{\text{dependency paths } P \rightarrow \dots \rightarrow Q} \prod_{\text{edges}} B$$

counts occurrence-generating paths with multiplicity.

8.3 A formal telescoping identity for software

Let $q(H_{\text{sw}})$ be the free abelian group on package-version atoms. Define the formal local increment

$$\Delta(P) = P(D(P))^{-1}.$$

Then every software product satisfies

$$S = \prod_P \Delta(P)^{M_P(S)} \quad \text{in } q(H_{\text{sw}}).$$

This is an exact cancellation identity. It is not an installation command and does not mean that dependencies are physically invertible; it is a provenance formula in the group completion.

8.4 Dependency signatures, kernels, and overlap

The vector

$$\Phi_{\text{sw}}(S) = M(S)$$

is a sparse transitive dependency signature. It supports

$$K(S, T) = \sum_P w_P M_P(S) M_P(T),$$

$$d(S, T)^2 = \sum_P w_P (M_P(S) - M_P(T))^2.$$

Possible weights include vulnerability severity, license risk, maintenance cost, or inverse package frequency.

The overlap product is defined by

$$M(S \star_D T) = M(S) \circ M(T).$$

Its support is exactly the intersection of recursive package supports:

$$\text{supp } M(S \star_D T) = \text{supp } M(S) \cap \text{supp } M(T).$$

It amplifies packages that occur frequently in both products. The boundary

$$(I - B)(M(S) \circ M(T))$$

produces a valid formal root-product vector because the recursive cone is Hadamard-closed.

8.5 Checkpoint fronts and control polynomials

Define the fully marked checkpoint polynomial of a package by

$$\Psi_T(z) = z_T \quad \text{for a terminal package } T,$$

$$\Psi_P(z) = z_P + \prod_Q \Psi_Q(z)^{B_{Q,P}}$$

for a nonterminal package P .

A monomial corresponds to a set of package occurrences that meets every root-to-terminal dependency path exactly once and never selects an ancestor together with its descendant. Such a front can represent:

- one audit checkpoint per dependency path;
- one vulnerability-inspection layer;
- one license-review layer;
- one caching or verification frontier.

The degree identity

$$\deg_{z_Q} \Psi_P = C_{Q,P}$$

recovers deterministic dependency capacities from the front polynomial.

If each checkpoint P has cost c_P , the minimum-cost complete checkpoint front satisfies the dynamic recursion

$$V(P) = \min \left\{ c_P, \sum_Q B_{Q,P} V(Q) \right\}.$$

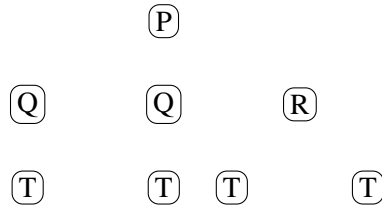
This is a tropical/min-plus realization of the same root-versus-descendants decomposition.

8.6 Worked software example

Consider package types T, Q, R, P with

$$D(T) = 1, \quad D(Q) = T, \quad D(R) = T^2, \quad D(P) = Q^2 R.$$

The unfolded trees are shown schematically below.



In the order (T, Q, R, P) ,

$$M(P) = (4, 2, 1, 1)^T.$$

The boundary gives

$$(I - B)M(P) = (0, 0, 0, 1)^T,$$

recovering one root occurrence of P and no other roots.

The fully marked checkpoint polynomials are

$$\Psi_T = z_T, \quad \Psi_Q = z_Q + z_T, \quad \Psi_R = z_R + z_T^2,$$

$$\Psi_P = z_P + (z_Q + z_T)^2 (z_R + z_T^2).$$

The terminal-count specialization $z_T = x$ and $z_P = z_Q = z_R = 1$ is

$$P_P(x) = 1 + (1 + x)^2 (1 + x^2).$$

Hence

$$P_P(1) = 9$$

complete checkpoint fronts exist, and

$$\deg P_P = 4 = M_T(P).$$

8.7 DAGs versus unfolded forests

A real dependency structure is usually a DAG with shared nodes. The recursive forest is its occurrence-unfolding.

- A package that appears once as a DAG node can occur many times in the unfolded forest.
- $M_Q(S)$ counts unfolded occurrences, equivalently weighted dependency paths, not necessarily distinct installed package records.
- Deduplicated component counts are different invariants and are not recovered by the linear occurrence theory alone.

This distinction must be stated in every empirical application.

8.8 Limits and required extensions

The finite theory assumes deterministic, acyclic, integer-multiplicity dependencies. Real systems may contain:

- version alternatives and optional dependencies;
- platform-conditioned edges;
- peer dependencies and conflicts;
- cycles;
- deduplication and shared runtime instances.

Cycles can be condensed into strongly connected components, but a component then ceases to be a free atom unless its internal behavior is modeled separately. Alternatives lead naturally to AND–OR trees, semiring-valued edge rules, or families of transition matrices.

9 Other hierarchical applications

The same axioms apply whenever objects have deterministic, finite, rank-decreasing decompositions into lower-level atom types.

Domain	Atoms and predecessor	Main Pratt-type outputs
Bill of materials	Part-version atoms; $D(P)$ is the multiset of direct subparts	BOM explosion $M = (I - B)^{-1}u$, provenance inversion, consistency cone, shared-part overlap
Attack or fault trees	Event or control types; children are direct subevents	Complete control fronts, weighted cut conservation, front enumerators, minimum-cost front recursion
Skill hierarchies	Skill atoms; $D(S)$ lists prerequisite subskills	Transitive skill signatures, root-skill recovery, kernels, overlap matching
Data provenance	Transformation atoms; $D(T)$ lists direct source transformations	Occurrence-path counts, origin inversion, consistency diagnostics

Domain	Atoms and predecessor	Main Pratt-type outputs
Hierarchical flows	work- Task types; $D(T)$ lists deterministic subtasks	Front checkpoints, probability laws, cumulants, workload bounds

10 What is genuinely universal, and what is not

10.1 Universal under the multiplicative axioms

The following require only a ranked recursive factorization system:

- finite trees and forests;
- $C = (I - B)^{-1}$ and weighted path counts;
- $M = Cv$ and boundary inversion $v = (I - B)M$;
- intrinsic cone characterization;
- quotient-group reconstruction;
- Hilbert embeddings, kernels, and metrics;
- Hadamard closure and the overlap semiring.

10.2 Universal for finite rooted forests

The following require only the forest, not factorization:

- complete fronts and maximal antichains;
- root-versus-descendants recursion;
- front-count and front-size inequalities;
- conserved-flow and Kraft identities;
- probability distributions on fronts;
- dynamic programming for minimum-cost fronts.

10.3 Realization-dependent

The following need additional algebraic structure:

- weighted complete-front expansions require addition in a semiring or ring;
- logarithmic differentiation and Mason–Stothers require a differential domain such as $\mathbb{Q}[x]$;
- numerical logarithmic inequalities require a positive evaluation compatible with the recursion;
- Galois and monodromy questions require a polynomial or function-field realization.

This separation prevents analogies from being mistaken for theorems.

11 Research program

Problem 11.1 (Polynomial Hadamard cone). Characterize the image of polynomial Pratt coordinates intrinsically and determine whether it is Hadamard-closed. The numerical proof suggests a positive answer whenever the transition multiplicities are nonnegative integers and boundary nonnegativity characterizes the image.

Problem 11.2 (Recursive UFDs). Classify factorial domains R admitting a normalized atom set $\mathcal{A}$, a rank-decreasing predecessor $D\alpha$, and an additive residual c_α with

$$\alpha = c_\alpha + D\alpha.$$

Determine which such systems have a distinguished terminal atom whose coordinate is a natural degree or capacity.

Problem 11.3 (DAG quotient theory). Develop coordinates that count distinct DAG nodes rather than unfolded occurrences while retaining a useful boundary inversion. Determine which nonlinear corrections replace $M = (I - B)^{-1}v$ under deduplication.

Problem 11.4 (AND–OR recursive systems). Extend the theory from deterministic products of children to alternative dependency clauses. A natural target is a semiring in which AND is multiplication and OR is addition or minimum.

Problem 11.5 (Software pilot). On a public package ecosystem, compare ordinary package-set similarity, graph embeddings, and recursive-coordinate kernels for predicting shared vulnerability exposure, upgrade impact, and dependency anomalies. Report separately occurrence-unfolded and deduplicated results.

Problem 11.6 (Front robustness). For dependency, attack, or fault trees, compare the minimum-cost front with the full front distribution. Quantify how many near-optimal fronts exist and how sensitive the optimum is to node failures or cost perturbations.

12 Conclusion

The common starting point is not a particular collection of primes or polynomials. It is a predecessor rule on atoms:

$$D\alpha = \prod_{\beta} \beta^{B_{\beta,\alpha}}, \quad \rho(\beta) < \rho(\alpha) \text{ whenever } B_{\beta,\alpha} > 0.$$

This rule generates finite atomic trees. Factorization of a general object supplies the roots of a forest. The full recursive population M and the root population v are related by

$$M = (I - B)^{-1}v, \quad v = (I - B)M.$$

The local increments telescope:

$$h = \prod_{\alpha} (\alpha/D\alpha)^{M_{\alpha}(h)}.$$

The natural coordinate cone is closed under coordinatewise multiplication, producing an overlap semiring. If the system also has an additive realization

$$g_\alpha = c_\alpha + \prod_{\beta} g_\beta^{B_{\beta,\alpha}},$$

then every realized object is a weighted sum over complete fronts. Front polynomials encode exact distributions, derivatives encode moments and covariances, and conserved weights cross every front unchanged.

Natural numbers, prime-index Pratt polynomials, and general monic rational polynomials are therefore not three unrelated constructions. They are three realizations of one recursive factorization architecture. Software dependency systems fit the same multiplicative-combinatorial architecture once package versions are treated as atoms and direct dependencies as predecessors. The resulting theory supplies exact provenance inversion, consistency constraints, sparse dependency signatures, overlap products, and complete checkpoint fronts, while also making explicit where real software requires extensions beyond deterministic acyclic unfolding.

A Dictionary of the principal specializations

Concept	Natural numbers	General monic polynomials	Software packages
Atom	prime p	irreducible φ	package-version P
Predecessor	$p - 1$	$\varphi - \varphi(0)$, with $Dx = 1$	direct dependency multiset $D(P)$
Terminal atom	2	x	package with no dependencies
Rank	prime size/order	degree	topological depth/order
Immediate-child matrix	$v_q(p - 1)$	$v_\psi(D\varphi)$	dependency multiplicity $B_{Q,P}$
Root vector	$v_p(n)$	$v_\varphi(f)$	declared root packages
Recursive vector	$m_p(n)$	$M_\varphi(f)$	unfolded occurrence counts
Boundary	ordinary factorization	ordinary irreducible factorization	root-package reconstruction
Local increment	$p/(p - 1)$	$\varphi/D\varphi$	formal $P/D(P)$
Additive realization	$p = 1 + (p - 1)$	$\varphi = \varphi(0) + D\varphi$	chosen counting/cost/risk semiring
Front polynomial	$f_n(x)$	weighted by constants and x	checkpoint polynomial Ψ_S

B Source notes used in the synthesis

The present text synthesizes and abstracts formulas and proofs from the following uploaded manuscripts:

1. O. Leka, *Pratt Coordinates for Natural Numbers and Monic Rational Polynomials: Reconstruction, Hilbert-Space Embeddings, and a Pratt-Coordinate Form of Mason–Stothers*, Version 0.2, August 3, 2026.
2. O. Leka, *Complete Node Fronts of Pratt Forests: Generating Polynomials and a Probabilistic Interpretation of $f_n(x)$* , August 5, 2026.
3. O. Leka, *Hadamard–Pratt Arithmetic: A Coordinatewise Product on Pratt Coordinates, a Semiring on $\mathbb{N}$, and the Spectrum of Its Unitization*, August 4, 2026.
4. O. Leka, *Root-Label Fields on Pratt Forests: An Application-First Multivariate Theory of Complete Fronts, Coordinate Transport, Arithmetic Fibres, and Monodromy*, Version 0.3, August 6, 2026.